Browser Forensics: DFIR Analysis of Web Browser Artifacts using Browser History Examiner

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Introduction

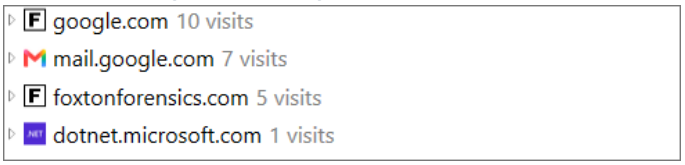
A web browser records much of what a person does online. For a Digital Forensics and Incident Response (DFIR) analyst these records answer who used the machine, what they visited, searched, downloaded and viewed, and when. This report examines fourteen browser artifact categories taken from my own laptop and explains what each shows and why it matters to an investigation.

Method and Tools

Tool: Foxton Forensics Browser History Examiner (BHE). **Sources:** Google Chrome (Default, Profile 1, 3 and 4) and Microsoft Edge (Default) on my own laptop. **Settings:** time zone UTC, date format dd/mm/yyyy. **Process:** browser data was captured into BHE, each artifact tab was opened and screenshotted, and each screenshot was cropped to the relevant columns. **Privacy:** email addresses, usernames, form values, saved-login URLs, page titles and faces were blurred. Record counts in each heading are the totals shown in BHE's artifact list.

Artifact Findings (1 of 2)

1. Website Visits (10,894 records)



Location: Chrome/Edge *History* database (urls and visits tables). **Observed:** 25 visits on 29/09/2026 at 09:00 UTC: google.com (10), mail.google.com (7), foxtonforensics.com (5), dotnet.microsoft.com (1). **DFIR value:** Backbone of the timeline. Shows which sites were visited, when, how often and in which profile, so activity can be tied to a user session.

2. Searches (4,029 records)

Date Searched	Search Terms	Search Engine	Source
29/09/2026 09:11:12	browser history examiner	Google	Chrome History
29/09/2026 09:11:12	browser history examiner	Google	Website Visit
29/09/2026 09:07:16	how do i know if my compu	Google	Website Visit
29/09/2026 09:07:12	how do i know if my compu	Google	Website Visit
29/09/2026 09:07:12	how do i know if my compu	Google	Website Visit
29/09/2026 09:05:23	how do i know if my compu	Google	Website Visit
29/09/2026 09:05:22	how do i know if my compu	Google	Website Visit
29/09/2026 09:05:22	how do i know if my compu	Google	Website Visit
29/09/2026 09:05:16	how do i know if my compu	Google	Website Visit
29/09/2026 09:05:15			Website Visit

Location: *History* database (search terms and query strings in URLs). **Observed:** Google queries on 29/09/2026 09:05-09:11 UTC, e.g. a question about the Microsoft .NET Framework, then "browser history examiner". **DFIR value:** Reveals intent, knowledge and planning. What a user searched just before an action (a download, a deletion) explains why the action happened.

3. Downloads (199 records)

State	Start Time	End Time	Bytes Downloaded
Complete	29/09/2026 09:16:42	29/09/2026 09:17:01	3462314
Complete	29/09/2026 09:06:08	29/09/2026 09:06:10	1439328
Complete	29/09/2026 06:24:10	29/09/2026 06:24:30	29020519
Complete	29/09/2026 00:41:10	29/09/2026 00:41:36	585715954
Complete	29/09/2026 00:39:43	29/09/2026 00:39:54	6519800
Complete	29/09/2026 00:36:33	29/09/2026 00:36:40	2269644
Cancelled	29/09/2026 00:18:45		0
Complete	29/09/2026 00:15:40	29/09/2026 00:17:13	212213456
Complete	28/09/2026 23:46:56	28/09/2026 23:49:58	1344208316
Complete	28/09/2026 23:30:11	28/09/2026 23:31:01	29020519

Location: *History* database (downloads and download chains tables). **Observed:** State, start/end time and size per file, e.g. a 3,462,314-byte ZIP completed 09:16:42-09:17:01 UTC (Chrome Profile 1); one download cancelled at 0 bytes. **DFIR value:** Shows malware delivery, tool staging and data movement. Source URL, target path, size and time link a file on disk to the site it came from. Cancelled downloads still prove intent.

4. Bookmarks (229 records)

Date Added	Title	URL
29/09/2022 22:48:30	New Tab	chrome://newtab/
23/09/2023 14:40:29	Morelivetv - Football Live Streaming, Watch Liv	https://afr.morelivet
23/09/2023 14:40:29	Morelivetv - Football Live Streaming, Watch Liv	https://afr.morelivet
13/05/2026 07:17:55	What is Claude?	https://anthropic.ski
23/05/2026 08:57:21	High-income jobs for university graduates - Cl	https://claude.ai/cha
02/06/2026 14:21:07	Messaging Excellence | Corporate Training with	https://copyhackers.
02/06/2026 14:24:38	Write Better Headlines: Headline Analyzer Fron	https://coschedule.c
04/04/2023 10:57:34	Internet Speed Test | Fast.com	https://fast.com/
04/04/2023 10:57:34	Internet Speed Test | Fast.com	https://fast.com/
02/06/2026 14:24:03	Hemingway Editor	https://hemingway.a

Location: Chrome *Bookmarks* JSON file and the Edge equivalent. **Observed:** Title, URL and date added per browser profile, from Edge entries dated 2022 to entries added on 29/09/2026. **DFIR value:** Shows sites the user considered important or revisited (interests, work tools, illicit resources). Survives history clearing, and the date added anchors the timeline.

5. Cookies (2,219 records)

Date Created	URL	Date Expires	Name
29/09/2026 09:32:28	.quillbot.com/	29/09/2026 10:02:28	__cf_bm
29/09/2026 09:32:14	.bing.com/	30/09/2026 09:32:14	.MSA.Auth
29/09/2026 09:32:14	.bing.com/		WLS
29/09/2026 09:32:14	.bing.com/		_EDGE_S
29/09/2026 09:32:14	.bing.com/		_SS
29/09/2026 09:32:14	.bing.com/	13/10/2026 09:32:14	_U
29/09/2026 09:32:14	.msn.com/	06/10/2026 09:32:14	elt
29/09/2026 09:32:14	.msn.com/	06/10/2026 09:32:14	eltc
29/09/2026 09:32:14	.msn.com/	29/10/2026 09:32:14	lt
29/09/2026 09:25:16	.google.com/	29/09/2027 09:25:16	SIDCC

Location: Chrome/Edge *Network\Cookies* database. **Observed:** Domain, created and expiry dates and cookie names, e.g. bing.com and msn.com cookies created 09:32:14 UTC (Edge) and Gmail/Google cookies. Values are empty because they are encrypted. **DFIR value:** Prove a site was visited even if history is wiped, and show first and last contact and login state. Session cookies can be stolen for session hijacking; tracking cookies profile browsing habits.

6. Logins (550 records)

Hostname	Origin URL	Submit URL	Username	Date Created
https://www.udemy.c				15/10/2025 19:08:32
https://www.pizzahut.				15/10/2025 18:51:04
https://www.jvzoo.cor				15/10/2025 18:58:46
https://www.gate.com				15/10/2025 18:35:17
https://www.football.(				17/03/2026 16:21:47
https://www.football.(				28/10/2025 06:57:41
https://www.credly.co				29/01/2026 23:00:02
https://swiftdrop-gh.l				29/05/2026 12:42:51
https://preview--swift				25/05/2026 12:44:54
https://lovable.dev/				23/05/2026 14:11:00

Location: *Login Data* database (saved credentials). Usernames and URLs blurred. **Observed:** Hostname and date created for saved accounts, including android:// entries. Passwords are stored encrypted and are not shown. **DFIR value:** Identifies accounts the user owns and when they were created. A prime target for infostealer malware. android:// entries suggest credentials synced from a phone.

Artifact Findings (2 of 2)

7. Email Addresses (308 records)

	Last Used		Domain	Source
	29/09/2026 09:16:32		mail.google.com	Website Visit
	29/09/2026 09:12:26			Form History
	26/09/2026 20:17:28	email addresses	teams.live.com	Website Visit
	26/09/2026 20:12:53		login.live.com	Website Visit
	07/09/2026 19:28:59		whop.com	Website Visit
	01/09/2026 18:32:03		consulting.com	Website Visit
	01/09/2026 18:30:32		event.monetise.com	Website Visit
	26/08/2026 08:39:37		accounts.google.com	Website Visit
	26/08/2026 08:39:37	BLURRED	google.com	Website Visit
	29/07/2026 19:30:22		login.microsoftonline.com	Website Visit

Location: Extracted by the tool from URLs, autofill and form data. Addresses blurred. **Observed:** Last used time, domain and source (Website Visit or Form History), e.g. Google, Microsoft, Teams and other services. **DFIR value:** Attributes activity to an identity, exposes aliases and other accounts, and gives investigators addresses to pivot on (mailbox review, preservation requests).

8. Form History (77 records)

Field Name	Value	First Used	Last Used	Times Used
ctl00$MainCo		29/09/2026 09:12:26	29/09/2026 09:12:26	1
ctl00$MainCo		29/09/2026 09:12:26	29/09/2026 09:12:26	1
q		07/09/2026 17:06:17	07/09/2026 17:06:17	1
q		07/09/2026 17:05:55	07/09/2026 17:05:55	1
order-numbe		27/06/2026 16:25:53	27/06/2026 16:26:02	2
query		03/06/2026 07:22:37	03/06/2026 07:22:37	1
query		03/06/2026 07:22:14	03/06/2026 07:22:14	1
query		03/06/2026 07:22:00	03/06/2026 07:22:00	1
q		03/06/2026 07:21:11	03/06/2026 07:21:11	1
q		03/06/2026 07:20:33	03/06/2026 07:20:33	1

Location: *Web Data* database (autofill table). Values blurred. **Observed:** Field name, first/last used and times used, e.g. email, tel, given-name, order-number. One email field was used 32 times, first used in 2020. **DFIR value:** Shows what the user typed into forms (names, phones, addresses, search boxes). Use counts and dates show how long and how often a detail was used, tying a person to an account.

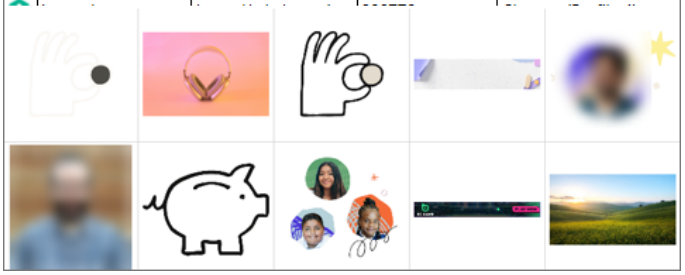
9. Thumbnails (19 records)

	URL	Title	Web Browser (Profile)
	https://chrome.google.com/w	Web Store	Chrome (Default)
	https://www.youtube.com/live	Side Hustle Summit | VIP SESSIO	Chrome (Default)
	chrome-extension://ekmeppjc	OrangeMonkey	Chrome (Default)
	http://fast.com/	Internet Speed Test | Fast.com	Chrome (Default)
	https://developer.microsoft.cc	Microsoft Edge WebView2 | Micr	Chrome (Default)
	https://chrome.google.com/w	Web Store	Chrome (Profile 1)
	https://whop.com/shs-vip-tick	SHS VIP Ticket | Whop	Chrome (Profile 1)
	https://en1.savefrom.net/2ID/	Download and install SaveFrom.r	Chrome (Profile 1)
	https://www.youtube.com/	YouTube	Chrome (Profile 1)
	https://chrome.google.com/w	Web Store	Chrome (Profile 3)
			Chrome (Profile 3)
			Chrome (Profile 3)

Location: *Top Sites* database and thumbnail store. Two work-related rows blurred. **Observed:** URL, title and browser profile of frequently visited pages, e.g. Web Store, YouTube, Fast.com and an event ticket page. **DFIR value:** Gives visual and title evidence of pages the user often opened, even after history is cleared. May expose internal or work pages.

10. Cached Images (4,111 records)

	Content Type	URL	File Size (Bytes)	Web Browser (Profile)
	image/gif	https://platform.clau	2405570	Chrome (Profile 4)
	image/jpeg	https://claude.ai/ima	2301480	Chrome (Profile 4)
	image/gif	https://platform.clau	1356143	Chrome (Profile 4)
	image/jpeg	https://tflteams.publi	1267978	Chrome (Profile 1)



Location: Browser *Cache_Data* folder. Faces of people blurred. **Observed:** Content type, source URL, size and recovered image previews (GIF, JPEG, PNG, AVIF, WebP). **DFIR value:** Rebuilds what was actually displayed on screen. Proves viewing, not just visiting, and can recover images from pages that are now deleted.

11. Favicons (974 records)

	Page URL	Last Updated	Web Browser (Profile)
	https://mail.google.com/mail/	29/09/2026 09:16:42	Chrome (Profile 1)
	https://mail.google.com/mail/	29/09/2026 09:16:42	Chrome (Profile 1)
	https://mail.google.com/mail/	29/09/2026 09:16:42	Chrome (Profile 1)
	https://mail.google.com/mail/	29/09/2026 09:16:42	Chrome (Profile 1)
	https://mail.google.com/mail/	29/09/2026 09:16:42	Chrome (Profile 1)
	https://mail.google.com/mail/	29/09/2026 09:16:42	Chrome (Profile 1)
	https://mail.google.com/mail/	29/09/2026 09:16:42	Chrome (Profile 1)
	https://mail.google.com/mail/	29/09/2026 09:16:42	Chrome (Profile 1)
	https://mail.google.com/mail/	29/09/2026 09:16:42	Chrome (Profile 1)
	https://mail.google.com/mail/	29/09/2026 09:16:42	Chrome (Profile 1)

Location: *Favicons* database (icon-to-page mapping). **Observed:** Page URL and last updated time, e.g. Gmail icons at 09:16:42, dotnet.microsoft.com at 09:04:58 and freeCodeCamp at 00:44:48 UTC. **DFIR value:** The mapping often outlives deleted history, so it proves a site was visited, and the update time gives an approximate visit time.

12. Session Tabs (1,256 records)

	Last Visited	Title	URL	Referrer	Web Browser (Profile)
	29/09/2026 09:16:45		https://www.fox	https://www.fox	Chrome (Profile 1)
	29/09/2026 09:16:38		https://www.fox		Chrome (Profile 1)
	29/09/2026 09:16:38	Foxton Forensics	https://www.fox		Chrome (Profile 1)
	29/09/2026 09:16:32	Browser History	https://mail.goo		Chrome (Profile 1)
	29/09/2026 09:16:32	Browser History	https://mail.goo		Chrome (Profile 1)
	29/09/2026 09:16:32	Browser Histor	https://mail.goo		Chrome (Profile 1)
	29/09/2026 09:16:25	Inbox (3,902)	https://mail.goo		Chrome (Profile 1)
	29/09/2026 09:16:25	Inbox (3,902)	https://mail.goo		Chrome (Profile 1)
	29/09/2026 09:15:53	Try EA SPORT	https://mail.goo		Chrome (Profile 1)
	29/09/2026 09:15:53	Try EA SPORT	https://mail.goo		Chrome (Profile 1)

Location: *Sessions* folder (Session_ and Tabs_ files). Part of Gmail titles blurred. **Observed:** Last visited, title, URL and referrer. Gmail tabs show the inbox count dropping from 3,904 to 3,902 in 35 seconds (09:15:50 to 09:16:25). **DFIR value:** Reconstructs what was open at the last close or crash. The referrer shows the navigation chain, and titles can reveal document or mailbox activity.

13. Cached Files (11,827 records)

	Content Type	URL	File Size (Bytes)	Web Browser (Profi
	Application/msi	https://www.foxtonfor	29020519	Chrome (Default)
	application/octet-strea	https://dl.google.com/	14657942	Chrome (Profile 4)
	application/octet-strea	https://dl.google.com/	4217751	Chrome (Profile 4)
	application/wasm	https://www.gstatic.co	3879679	Chrome (Profile 4)
	application/zip	https://www.foxtonfor	3462314	Chrome (Profile 1)
	application/gzip	https://platform.claud	2587860	Chrome (Profile 4)
	application/pdf	https://dn720904.ca.ar	2269644	Chrome (Default)
	video/webm	https://claude.ai/imag	1915225	Chrome (Profile 4)
	text/javascript	https://www.youtube.c	1913893	Chrome (Profile 4)
	text/javascript	https://assets-proxy.ar	1858011	Chrome (Profile 4)

Location: Browser *Cache_Data* folder. **Observed:** Content type, URL and size, e.g. an MSI installer (29,020,519 bytes) and a ZIP (3,462,314 bytes) from foxtonforensics.com, plus JavaScript and PDF files. **DFIR value:** May hold the real content served to the browser (executables, scripts, documents), so payloads can be recovered. Sizes can be matched to Downloads entries.

14. Cached Web Pages (394 records)

	URL	Fetch Count	File Size (Bytes)	Web Browser (Profile)
	https://www.gstatic.com/video_effects/ass		673008	Chrome (Profile 4)
	https://www.google.com/search?rlz=1C1R		589453	Chrome (Profile 4)
	https://www.google.com/search?q=sourci		586417	Chrome (Profile 4)
	https://www.google.com/search?q=contra		530807	Chrome (Profile 4)
	https://www.google.com/search?q=public		519594	Chrome (Profile 4)
	https://www.google.com/search?q=can+c		510472	Chrome (Profile 3)
	https://www.google.com/search?gs_ssp=e		501828	Chrome (Profile 4)
	https://www.google.com/search?q=p&rlz=		499818	Chrome (Profile 3)
	https://www.google.com/search?q=lovabl		478485	Chrome (Profile 4)
	https://www.google.com/search?q=how+		452307	Chrome (Profile 4)
	https://www.google.com/search?q=whats		441222	Chrome (Profile 4)
	https://www.google.com/search?q=googl		437053	Chrome (Profile 4)

Location: Browser *Cache_Data* folder. **Observed:** Mostly Google search result pages (google.com/search?q=...) with size and profile (Chrome Profile 3 and 4). **DFIR value:** Recovers pages as the user saw them, including embedded queries. Can preserve web content (mail, chat, forms) after it is deleted online.

How the Artifacts Combine: Timeline from this Laptop

No single artifact tells the whole story. Correlating them across profiles reconstructs my own activity on 28-29 September 2026 (times in UTC):

Time	Event	Artifacts that corroborate it
28/09 23:30	29,020,519-byte MSI installer downloaded from foxtonforensics.com (Chrome Default)	Downloads, Cached Files (same size)
29/09 00:18	Second download of the same installer cancelled at 0 bytes	Downloads (state: Cancelled)
29/09 06:24	Same 29,020,519-byte installer downloaded again	Downloads, Cached Files
29/09 09:04-09:07	Visit to dotnet.microsoft.com, then Google searches about the .NET Framework (Profile 1)	Favicons, Website Visits, Searches
29/09 09:06	1,439,328-byte file downloaded (Chrome Default) and cached from a Microsoft download host	Downloads, Cached Files (same size)
29/09 09:11	Search "browser history examiner"; foxtonforensics.com tab open	Searches, Session Tabs
29/09 09:15-09:17	Gmail tabs open (unread count 3,904 to 3,902); 3,462,314-byte ZIP downloaded (Profile 1)	Session Tabs, Cookies, Favicons, Downloads, Cached Files
29/09 09:32	Edge creates bing.com and msn.com cookies	Cookies

Two points stand out. First, matching file sizes between Downloads and Cached Files prove that a file named in the download list was really fetched, and the cache may still hold its content. Second, activity in the same minutes across different profiles (Default and Profile 1) shows the profile split alone does not separate users, so timing, accounts and cookies must be compared to attribute activity to a person.

Why This Matters to a DFIR Analyst

• **Timeline and attribution:** visits, searches, session tabs and cookies show what happened and when; logins, email addresses and form history show who.
• **Intent and knowledge:** searches, bookmarks and cancelled downloads show what the user was trying to do, even when the action failed.
• **Evidence of malware and data theft:** downloads, cached files and login data show payload delivery, infostealer targets and exfiltration paths.
• **Persistence after deletion:** favicons, thumbnails, cache and cookies often survive after history is cleared, so a deleted trail can be rebuilt.
• **Content recovery:** cached pages and images recreate what the user actually saw on screen.

Limitations

• Private/Incognito browsing is not written to disk, and users can clear history, cache and cookies, so absence of data is not proof of absence of activity.
• Cookie values and saved passwords are encrypted, so BHE shows names, dates and hosts but not the secrets themselves.
• Synced data (for example android:// logins) can appear on this laptop although it originated on another device.
• Timestamps depend on the system clock and time zone, and caches are overwritten over time.
• The BHE view here shows 25 records per artifact (likely the trial limit), so screenshots are a sample of the totals listed in the artifact panel. Some columns were truncated.
• Blurring hides identifying values, so those columns cannot be verified from the screenshots alone.

Conclusion

Each browser artifact answers a different investigative question, and together they form a strong, cross-checkable picture of user activity. Browser History Examiner makes this practical by parsing Chrome and Edge into one timeline. A DFIR analyst should preserve the profile folders, work on a copy, record the time zone, and correlate artifacts (as done above with download sizes and timestamps) before drawing conclusions.